

Proper Title

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Abstract

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Keywords: Keyords go here.

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1 Introduction

2 Problem formulation

2.1 Finite-strain elastostatics as energy minimisation

We consider a hyperelastic body occupying a reference domain $\Omega \subset \mathbb{R}^2$ with boundary $\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}$, $\Gamma_D \cap \Gamma_N = \emptyset$. The deformation of the body is described by the displacement field $u : \Omega \rightarrow \mathbb{R}^2$, the deformation map $y(x) = x + u(x)$, and the associated deformation gradient

$$F = \nabla y = I + \nabla u, \quad J = \det F > 0, \quad (1)$$

where ∇ denotes the gradient with respect to the reference configuration. The mechanical response is encoded in a strain energy density $\Psi(F)$, and the equilibrium configuration is the minimiser of the total potential energy

$$\mathcal{E}(u) = \int_{\Omega} \Psi(F) dV - \int_{\Omega} f_0 \cdot u dV - \int_{\Gamma_N} t_0 \cdot u dS, \quad (2)$$

over the space of admissible displacements $U = \{u \in H^1(\Omega; \mathbb{R}^2) : u = g \text{ on } \Gamma_D\}$, where f_0 is a body force, t_0 a prescribed traction on the Neumann boundary, and g the prescribed boundary displacement. We seek $u \in U$ such that $\mathcal{E}(u) \leq \mathcal{E}(v)$ for all admissible v .

2.2 Stationarity, stress, and tangent modulus

A minimiser satisfies the stationarity condition $D_{\delta u} \mathcal{E}(u) = 0$ for all admissible variations δu (vanishing on Γ_D), which is the weak form of equilibrium

$$\int_{\Omega} P(F) : \nabla \delta u dV = \int_{\Omega} f_0 \cdot \delta u dV + \int_{\Gamma_N} t_0 \cdot \delta u dS, \quad (3)$$

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where the first Piola–Kirchhoff stress is the first derivative of the energy with respect to the deformation gradient,

$$P(F) = \frac{\partial \Psi}{\partial F}. \quad (4)$$

The natural boundary condition associated with Γ_N is $PN = t_0$, with N the reference outward unit normal. Because (3) is nonlinear in u , it is solved iteratively by Newton’s method. Linearising the residual about a current iterate $\bar{u}$ in a direction Δu introduces the fourth-order tangent (elasticity) modulus

$$L = \frac{\partial P}{\partial F} = \frac{\partial^2 \Psi}{\partial F \otimes \partial F}, \quad (5)$$

so that each Newton step requires solving the linear problem

$$\int_{\Omega} \nabla \delta u : L(\bar{F}) : \nabla \Delta u \, dV = -R(\bar{u}; \delta u), \quad (6)$$

with $R(\bar{u}; \delta u)$ the residual of (3) evaluated at $\bar{u}$. The well-posedness and conditioning of (6)—and hence the convergence of the whole nonlinear solver—are governed by the coercivity and boundedness of the bilinear form induced by $L(\bar{F})$. The central object of this work is therefore the behaviour of L as a function of the local deformation state.

2.3 The neo-Hookean material

Throughout we use the compressible neo-Hookean strain energy density adopted in the cut-finite-element study [?] that motivates this analysis,

$$\Psi(F) = \frac{\mu}{2} (\text{tr}(F^T F) - 2) + f(J), \quad (7)$$

where $\mu > 0$ is the shear modulus and $f(J)$ is a strictly convex volumetric penalty with $f(1) = 0$, $f'(1) = -\mu$, and $f(J) \rightarrow \infty$ as $J \rightarrow 0^+$ (for instance $f(J) = -\mu \log J + \frac{\lambda}{2} \log^2 J$, or $f(J) = \frac{\kappa}{2}(J - 1)^2$). The corresponding stress and tangent are

$$P(F) = \mu F + f'(J) J F^{-T}, \quad (8)$$

$$L_{ijkl} = \mu \delta_{ik} \delta_{jl} + (f''(J) J^2 + f'(J) J) F_{iL}^{-T} F_{kJ}^{-T} - f'(J) J F_{iJ}^{-T} F_{kL}^{-T}. \quad (9)$$

The first term $\mu \delta_{ik} \delta_{jl}$ is constant; the entire nonlinear part is controlled by J and the *inverse* deformation gradient F^{-1} , yielding the structural bound

$$\|L(F)\| \leq \mu + C(J) \|F^{-1}\|^2, \quad (10)$$

where $\|F^{-1}\| = \lambda_{\min}^{-1}$ is set by the smallest principal stretch. Consequently L can only become unbounded through volumetric collapse, $\lambda_{\min} \rightarrow 0$ (equivalently $J \rightarrow 0$); a purely tensile blow-up of F with F^{-1} bounded leaves L finite. This dichotomy is the analytical backbone of the present paper.

2.4 The mixed-corner singularity

The configuration of interest is a right-angle corner at the origin where the two boundary-condition types meet: a Dirichlet segment (Γ_D , prescribed displacement) abutting a traction-free Neumann segment (Γ_N). Such junctions arise, for example, at the lower corners of the loaded-pole benchmark

of [?], where the numerically observed convergence rate is capped well below the optimal order. Classical Kondratiev theory for the linearised problem predicts a corner singularity

$$u(r, \theta) \sim K r^\alpha \Phi(\theta), \quad \alpha \in (0, 1), \quad (11)$$

in polar coordinates (r, θ) centred at the corner, with K a generalised stress-intensity factor and α the (geometry- and Poisson-ratio-dependent) singularity exponent. The deformation gradient then scales as $F = I + \mathcal{O}(K r^{\alpha-1})$, so $\|F\| \rightarrow \infty$ as $r \rightarrow 0$.

The question this raises—and which the remainder of the paper addresses—is how this kinematic singularity propagates into the nonlinear tangent operator (5). In view of (10), the answer hinges on the *sign* of the loading (the sign of K): a tensile corner keeps F^{-1} bounded and leaves L well behaved, whereas a compressive corner drives $\lambda_{\min} \rightarrow 0$, blows up L , and triggers a loss of strong ellipticity. The following sections analyse these two scenarios in turn (Section 3) and then show how a small imperfection regularises the compressive case through a structural bifurcation (Section 4).

3 Asymptotic Behavior of the Neo-Hookean Operator at a Mixed Corner

3.1 Problem Setup and Linearized Kinematics

Let the reference domain $\Omega \subset \mathbb{R}^2$ feature a right-angle corner at the origin $(0, 0)$ described by polar coordinates (r, θ) for $r \in [0, R]$ and $\theta \in [0, \pi/2]$.

- **Dirichlet Boundary** (Γ_D): $\theta = 0$, where deformation is prescribed: $y(r, 0) = g(r)$.
- **Neumann Boundary** (Γ_N): $\theta = \pi/2$, which is traction-free: $PN = 0$.

Let $u(x)$ be the displacement, and $F = I + \nabla u$ the deformation gradient. From classical Kondratiev theory for linear elasticity, the leading-order singular displacement field at this mixed corner behaves as:

$$u(r, \theta) \sim K r^\alpha \Phi(\theta) \quad (12)$$

where K is the generalized stress intensity factor (dependent on far-field loading), $\Phi(\theta)$ is a smooth angular function, and $\alpha \in (0, 1)$ is the singularity exponent (typically $\alpha \approx 0.68$ for incompressible materials).

Consequently, the deformation gradient scales as:

$$F = I + \mathcal{O}(K r^{\alpha-1}) \quad (13)$$

As $r \rightarrow 0$, $\|F\| \rightarrow \infty$. The nature of this singularity—and the resulting behavior of the nonlinear tangent modulus $L = \frac{\partial^2 \Psi}{\partial F \partial F}$ —depends entirely on the sign of K (the loading direction).

We define a standard compressible neo-Hookean energy density:

$$\Psi(F) = \frac{\mu}{2} (\text{tr}(F^T F) - 2) + f(J) \quad (14)$$

where $J = \det(F)$ and $f(J)$ is a strictly convex volumetric penalty function satisfying $f(1) = 0$, $f'(1) = -\mu$, and $f(J) \rightarrow \infty$ as $J \rightarrow 0$.

The fourth-order tangent modulus L is exactly:

$$L_{iJkL} = \mu \delta_{ik} \delta_{JL} + \mathcal{B}_{iJkL}(F^{-1}, J) \quad (15)$$

where $\mathcal{B}$ contains all nonlinear geometric stiffness terms, bounded by $\|\mathcal{B}\| \leq C(J)\|F^{-1}\|^2$.

3.2 Scenario A: The Tensile (Non-Compressive) Singularity

Assume the far-field loading pulls the material away from the corner ($K > 0$ such that the material undergoes radial tension).

Kinematic consequence. The maximum principal stretch $\lambda_{\max}$ aligns roughly radially and blows up as $r^{\alpha-1}$. However, because the material is pulled outward, the domain locally expands or preserves volume (by drawing material in from the Neumann boundary). Therefore, the determinant J remains bounded strictly away from zero ($J \geq J_{\min} > 0$), and the transverse stretch $\lambda_{\min}$ remains finite and non-zero.

Operator behavior. Because $\lambda_{\min} > 0$, the inverse deformation gradient is bounded: $\|F^{-1}\| = \lambda_{\min}^{-1} \leq M$. Looking at the tangent modulus L :

$$\|L(x)\| \leq \mu + C(J)\|F^{-1}\|^2 < \infty \quad \text{as } r \rightarrow 0 \quad (16)$$

Conclusion for Scenario A. Although there is a severe *stress singularity* (the First Piola-Kirchhoff stress $P \sim \mu F \rightarrow \infty$ as $r \rightarrow 0$), **there is no singularity in the tangent operator.**

- L remains strongly elliptic (the Legendre–Hadamard condition $L_{iJkL}m_iM_Jm_kM_L \geq c > 0$ holds).
- The standard weak form of the linearized Newton step $\int_{\Omega} L : \nabla \delta u : \nabla \Delta u \, d\Omega$ remains uniformly coercive.
- The nonlinear neo-Hookean problem behaves exactly parallel to the linear problem. The singularity is safely handled by standard weighted Sobolev spaces ($V_{\beta}^{1,2}$), and no structural instability occurs.

3.3 Scenario B: The Compressive Singularity (The Pathological Case)

Now, assume the far-field loading pushes the material *into* the corner ($K < 0$, radial compression).

Kinematic consequence. The naive linear prediction suggests ∇u has a negative dominant eigenvalue scaling as $-|K|r^{\alpha-1}$. Because $F = I + \nabla u$, as we approach the corner ($r \rightarrow 0$) or as we increase the pseudo-time load factor t , there exists a critical threshold where the local contraction completely annihilates the material volume:

$$\det(F) \rightarrow 0 \implies J \rightarrow 0 \quad (17)$$

When $J \rightarrow 0$, the minimum principal stretch $\lambda_{\min} \rightarrow 0$.

Operator blow-up. Because $\|F^{-1}\| = \lambda_{\min}^{-1} \rightarrow \infty$, the nonlinear operator $\mathcal{B}$ dominates. The tangent modulus scales asymptotically as:

$$\|L(x)\| \sim \mathcal{O}(J^{-2}) \sim \mathcal{O}(\|F^{-1}\|^2) \rightarrow \infty \quad (18)$$

Loss of strong ellipticity. Before J even reaches 0, the extreme compression causes the acoustic tensor $Q_{ik}(N) = L_{iJkL}N_JN_L$ to lose positive-definiteness. Let N be the normal vector aligned with the direction of maximum compression. The acoustic tensor associated with this direction will develop a zero, and then negative, eigenvalue:

$$\min_{m,N} (L_{iJkL}m_iN_Jm_kN_L) \leq 0 \quad (19)$$

The PDE locally ceases to be elliptic and becomes hyperbolic.

Conclusion for Scenario B. The naive assumption that the deformation smoothly follows the fundamental thermodynamic path into the corner leads to a complete mathematical breakdown. The tangent modulus L becomes infinitely singular ($\|L\| \rightarrow \infty$), and the local operator loses coercivity, meaning standard Newton–Raphson solvers will inevitably fail, and a weak solution for the linearized step ceases to exist in the standard functional spaces.

This unphysical infinite accumulation of energy ($f(J) \rightarrow \infty$) and operator blow-up implies that the assumption of a smooth deformation path is flawed. In reality, the loss of ellipticity dictates a local geometric bifurcation. To capture this analytically, we must introduce an imperfect perturbation (noise) to regularize the bifurcation and bound the singularity.

4 Imperfect Bifurcation Analysis and Asymptotic Bounds

4.1 The Perturbed Weak Form

We introduce a pseudo-time load parameter $t \in [0, 1]$ which monotonically scales the compressive Dirichlet boundary conditions. To prevent the unphysical perfect fundamental path ($J \rightarrow 0$), we introduce a small, fixed volumetric perturbation (noise) $\epsilon f(x)$, where $\|f\|_{L^2} = 1$ and $\epsilon \ll 1$.

The nonlinear equilibrium equation in weak form is $G(u, t, \epsilon) = 0$, defined as:

$$\langle G(u, t, \epsilon), v \rangle = \int_{\Omega} P(F) : \nabla v \, d\Omega - \epsilon \int_{\Omega} f \cdot v \, d\Omega = 0 \quad \forall v \in V \quad (20)$$

where $P(F) = \frac{\partial \Psi}{\partial F}$ is the First Piola-Kirchhoff stress, and V is the appropriate space of kinematically admissible test functions.

4.2 The Critical Point (Perfect System, $\epsilon = 0$)

Let $u_0(t)$ be the fundamental path of pure compression (the unphysical path where $J \rightarrow 0$ as $t \rightarrow 1$). At some critical load $t_c < 1$, the tangent modulus $L(F_0)$ evaluated on the fundamental path loses positive definiteness. The linearized operator $\mathcal{L}_c := \partial_u G(u_0(t_c), t_c, 0)$ develops a zero eigenvalue.

Let $\phi \in V$ be the normalized critical eigenmode ($\|\phi\|_V = 1$) associated with this instability (representing a localized rotation/buckling near the corner). It satisfies:

$$\int_{\Omega} \nabla \phi : L_c : \nabla v \, d\Omega = 0 \quad \forall v \in V \quad (21)$$

4.3 Lyapunov–Schmidt Reduction (The Normal Form)

To analyze the perturbed system ($\epsilon > 0$) near t_c , we decompose the displacement field into the fundamental path, the buckling amplitude $w(t, \epsilon)$ along the eigenmode ϕ , and a higher-order orthogonal remainder v :

$$u(t, \epsilon) = u_0(t) + w\phi + v \quad (22)$$

By projecting the weak form onto the eigenmode ϕ and applying a standard Taylor expansion of the energy functional up to the fourth order in w , the infinite-dimensional PDE reduces to a scalar algebraic equation for the amplitude w (the imperfect bifurcation equation):

$$a(t_c - t)w + bw^3 = \epsilon\eta \quad (23)$$

where:

- $a = \frac{1}{2} \frac{d}{dt} \langle \phi, \partial_u^2 G(u_0(t_c)) \phi \rangle$ is the eigenvalue crossing velocity.
- $b = \frac{1}{6} \langle \phi, \partial_u^3 G(u_0(t_c)) [\phi, \phi, \phi] \rangle$ governs the post-buckling stiffness (assumed $b > 0$ for a stable supercritical bifurcation).
- $\eta = \int_{\Omega} f \cdot \phi \, d\Omega$ is the projection of the noise onto the buckling mode.

4.4 Asymptotic Estimate of the Bifurcation Amplitude

We are interested in the maximum deformation amplitude, which occurs at or near the critical time $t = t_c$. Evaluated exactly at t_c , the linear term vanishes, leaving:

$$bw^3 \approx \epsilon\eta \quad (24)$$

Thus, the amplitude of the structural instability scales asymptotically with the perturbation:

$$w(\epsilon) \sim \left(\frac{\eta}{b}\right)^{1/3} \epsilon^{1/3} \quad \text{as } \epsilon \rightarrow 0 \quad (25)$$

Remark 1. Even if the material softening dictates an asymmetric/transcritical bifurcation yielding $w^2 \sim \epsilon$, the logic below remains identical with a modified exponent. We proceed with the standard pitchfork $\epsilon^{1/3}$ scaling.

4.5 Kinematic Relief Mechanism: Bounding J and $\|F^{-1}\|$

The core physical mechanism is that the buckling mode ϕ introduces *rotation and shear*, which geometrically relieves the pure volumetric compression.

At the corner, the total deformation gradient is $F = F_0(t_c) + w\nabla\phi + \mathcal{O}(w^2)$.

- $F_0(t_c)$ is a purely compressive state with a minimum principal stretch λ_c , which is small but strictly non-zero (t_c occurs *before* $J \rightarrow 0$).
- The buckling gradient $\nabla\phi$ represents a symmetry-breaking rotation.

Because the instability trades pure compression for bending/rotation, the volumetric determinant $J = \det(F)$ becomes bounded away from zero by the amplitude of the buckling mode. To leading order, rotations modify the stretch quadratically. Therefore, the minimum volume ratio is bounded below by:

$$J_{\min}(\epsilon) \geq C_1 \lambda_c + C_2 w^2 \quad (26)$$

where $C_1, C_2 > 0$ are geometric constants. As $\epsilon \rightarrow 0$ (and assuming λ_c approaches some infinitesimal limit near the theoretical singularity), the dominant bound is provided by the buckling amplitude:

$$J_{\min}(\epsilon) \sim \mathcal{O}(w^2) \sim \mathcal{O}(\epsilon^{2/3}) \quad (27)$$

Since $J = \det(F) = \lambda_1 \lambda_2$ (in 2D), and the transverse stretch is largely unaffected or bounded, the minimum principal stretch scales as $\lambda_{\min} \sim \mathcal{O}(\epsilon^{2/3})$.

Consequently, the norm of the inverse deformation gradient is strictly bounded by the noise:

$$\|F^{-1}\| = \frac{1}{\lambda_{\min}} \sim \mathcal{O}(\epsilon^{-2/3}) \quad (28)$$

4.6 Final Asymptotic Bound on the Tangent Modulus L

Recall from the previous section that the norm of the fourth-order neo-Hookean tangent modulus L is bounded by the inverse deformation gradient:

$$\|L(F)\| \leq \mu + \mathcal{K}(J)\|F^{-1}\|^2 \quad (29)$$

Assuming the volumetric penalty $f(J)$ is chosen such that $\mathcal{K}(J)$ remains bounded or scales no worse than J^{-1} near $J = 0$, the dominant singularity is driven by $\|F^{-1}\|^2$.

Substituting the kinematic bound derived from the imperfect bifurcation analysis:

$$\max_{x \in \Omega} \|L(x, \epsilon)\| \leq C\|F^{-1}\|^2 \sim \mathcal{O}\left((\epsilon^{-2/3})^2\right) = \mathcal{O}(\epsilon^{-4/3}) \quad (30)$$

5 Conclusions

References